A
Mini Project Report
On

DETECTING SPAMMERS AND IDENTIFYING FAKE USERS ON SOCIAL NETWORKS

Submitted to

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for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE & ENGINEERING

SREE CHAITANYA
EDUCATIONAL INSTITUTIONS

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Certificate



Certified that this Mini Project Report entitled, "DETECTING SPAMMERS AND IDENTIFYING FAKE USERS ON SOCIAL NETWORKS" is the bonafide work of PEDDI MANIPRIYA(HTNO:23TR1A0589), MOHAMMAD AWAIS KHAN(HTNO: 23TR1591) ,PEDDAVENI RANADEEPIKA (HTNO: 23TR1A0586),KONDA ABHIRAM (HTNO: 23TR1A05B9)of III Year, CSE in the year 2026 in partial fulfilment of the requirements to award the Degree of Bachelor of Technology in Computer Science & Engineering Branch of Sree Chaitanya Institute of Technological Sciences, Karimnagar.

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Finally Special thanks to our parents, sisters and brothers for their support and encouragement throughout my life and this course. Thanks to all our friends and well-wishers for their constant support.

DECLARATION

We hereby declare that the work which is being presented in this report entitled, **DETECTING SPAMMERS AND IDENTIFYING FAKE USERS ON SOCIAL NETWORKS**,submitted towards the partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science Engineering**, Sree Chaitanya Institute of Technological Sciences, Karimnagar is an authentic record of our own work carried out under the supervision of **DR. PEDDI KISHOR, Professor & HOD, Department of CSE,** Sree Chaitanya Institute of Technological Sciences, Karimnagar.

To the best of our knowledge and belief, this project bears no resemblance with any report submitted to Sree Chaitanya Institute of Technological Sciences or any other University for the award of any degree or diploma.

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ABSTRACT

Social networking sites engage millions of users around the world. The user's interactions with these social sites, such as Twitter and Facebook have a tremendous impact and occasionally undesirable repercussions for daily life. The prominent social networking sites have turned into a target platform for the spammers to disperse a huge amount of irrelevant and deleterious information. Twitter, for example, has become one of the most extravagantly used platforms of all times and therefore allows an unreasonable amount of spam. Fake users send undesired tweets to users to promote services or websites that not only affect legitimate users but also disrupts our misconception. Moreover, the possibility of expanding in valid information to users through fake identities has increased that results in the unrolling of harmful content. Recently, the detection of spammers and identification of fake users on Twitter has become a common area of research in contemporary online social Networks (OSNs). In this paper, we perform a review of techniques used for detecting spammers on Twitter.Moreover, a taxonomy of the Twitter spam detection approaches is presented that classifies the techniques based on their ability to detect: (i) fake content, (ii) spam based on URL, (iii) spam in trending topics, and (iv) fake users. The presented techniques are also compared based on various features, such as user features, content features, Graph features, structure features, and time features. We are hopeful that the presented study will be a useful resource for researchers to find the highlights of recent developments in Twitter spam detection on a single platform.

LIST OF FIGURES

S.No No	Figure Description	Page
1.	Fig.3.1: block diagram	10
2.	Fig.3.2.1: level O data flow diagram	11
3.	Fig.3.2.2: level 1 data flow diagram	12
4.	Fig.3.2.3: level 2 data flow diagram	12
5.	Fig.3.2.4: Detection and Analysis system	13
6.	Fig.3.3: Actor-Driven Twitter Trend	15
7.	Fig.3.4: UML Class Diagram of the Twitter Data	16
8.	Fig.3.5: Sequence Diagram for Twitter Trend	17
9.	Fig.3.6: activity diagram for twitter trend detection	18
10.	Fig.3.7: Flowchart of Twitter Trend Analysis	19
11.	Fig.4.0: Java Program Compilation	21
12.	Fig.4.1: Platform Independence of Java Using JVM	21
13.	Fig.4.2: Components of the Java Platform Architecture	22
14.	Fig.4.3: Components of the Java 2 SDK Architecture	23
15.	Fig.4.4: process of java program execution	24
16.	Fig.4.5: TCP/IP stack	25
17.	Fig.4.6: a 32-bit IP address 137.92.11.1	27
18.	Fig.4.7: Initial Interface of Spammer and Fake User	37
19.	Fig.4.8: Uploading Tweets Data-set into the Spammer	37
20.	Fig.4.9: Tweets Dataset	38
21.	Fig.5.0: Detection Results Displayed After Analyzing	39

22. Fig.5.1: Tweet Data Displayed Before Running 40
23. Fig.5.2: Random Forest Classification 40
24. Fig.5.3: Random Forest Classification Results 41

TABLE OF CONTENTS

Abstract	V
List of Figures	VI-VII
List of Tables	VIII-IX

Page Nos Start – End

Chapter 1: INTRODUCTION 1-3

Chapter 2: SYSTEM ANALYSIS 4-9

2.1: LITERATURE REVIEW.....4-6

2.2: EXSISTING SYSTEM 7

2.3: PROPOSED SYSTEM 7

2.4: FEASIBILITY STUDY.....8-9

2.4.1: ECONOMICAL FEASIBILITY.....8

2.4.2: TECHNICAL FEASIBILITY.....8

2.4.3: SOCIAL FEASIBLITY.....9

2.5: SYSTEM REQUIREMENTS SPECIFICATION.....9

2.5.1: HARDWARE REQUIREMENTS.....9

2.5.2: SOFTWARE REQUIREMENTS.....9

Chapter 3: PROJECT DESIGN.....10-19

3.1: BLOCK DIAGRAM.....10

3.2: DATA FLOW DIAGRAM.....11-13

3.3: UML DIAGRAM.....14

3.4: USE CASE DIAGRAM.....15

3.5: CLASS DIAGRAM.....16

3.6: SEQUENCE DIAGRAM.....17

3.7: ACTIVITY DIAGRAM.....18

3.8: COLLABORATIVE DIAGRAM.....19

Chapter 4: IMPLEMENTATION.....20-36

Chapter 5: RESULTS.....37-41
Chapter 6: TESTING.....42-45
6.1: SYSTEM TESTING.....42
6.2: TYPES OF TESTING.....42-45
6.2.1: UNIT TESTING.....42
6.2.2: INTEGRATION TESTING.....43
6.2.3: FUNCTIONAL TESTING.....43
6.2.4: SYSTEM TESTING.....43
6.2.5: WHITEBOX TESTING.....44
6.2.6: BLACKBOX TESTING.....44
6.3: TEST STRATEGIE AND APPROACH.....45
Chapter 7: CONCLUSION.....46
REFERENCES.....47-48

CHAPTER-1
INTRODUCTION

It has become quite unpretentious to obtain any kind of information from any source across the world by using the Internet. The increased demand of social sites permits users to collect abundant amount of information and data about users. Huge volumes of data available on these sites also draw the attention of fake users . Twitter has rapidly become an online source for acquiring real-time information about users. Twitter is an Online Social Network (OSN) where users can share anything and everything, such as news, opinions, and even their moods. Several arguments can be held over different topics, such as politics, current affairs, and important events. When a user tweets something, it is instantly conveyed to his/her followers, allowing them to outspread the received information at a much broader level With the evolution of OSNs, the need to study and analyze users' behaviors in online social platforms has intensified. Many people who do not have much information regarding the OSNs can easily be tricked by the fraudsters. There is also a demand to combat and place a control on the people who use OSNs only for advertisements and thus spam other people's accounts.

Recently, the detection of spam in social networking sites attracted the attention of researchers. Spam detection is a difficult task in maintaining the security of social networks. It is essential to recognize spams in the OSN sites to save users from various kinds of malicious attacks and to preserve their security and privacy. These hazardous manoeuvres adopted by spammers cause massive destruction of the community in the real world. Twitter spammers have various objectives, such as spreading invalid information, fake news, rumours, and spontaneous messages. Spammers achieve their malicious objectives through

advertisements and several other means where they support different mailing lists and subsequently dispatch spam messages randomly to broadcast their interests. These activities cause disturbance to the original users who are known as non-spammers. addition, it also decreases the repute of the OSN platforms. Therefore, it is essential to design a scheme to spot spammers so thatcorrective efforts can be taken to counter their malicious activities.Several research works have been carried out in the domain of Twitter spam detection. To encompass the existing state-of the- art, a few surveys have also been carried out on fake user identification from Twitter. Tinman *et al.* provide a survey of new methods and techniques to identify Twitter spam detection. The above survey presents a comparative study of the current approaches. On the other hand, the authors in conducted a survey on different behaviours exhibited by spammers on Twitter social network.

The study also provides a literature review that recognizes the existence of spammers on Twitter social network. Despite all the existing studies, there is still a gap in the existing literature. Therefore, to bridge the gap, we review state-of-the-art in the spammer detection and fake user identification on Twitter. Moreover, this survey presents a taxonomy of the Twitter spam detection approaches and attempts to offer a detailed description of recent developments in the domain.

The aim of this paper is to identify different approaches of spam detection on Twitter and to present a taxonomy by classifying these approaches into several categories. For classification, we have identified four means of reporting spammers that can be helpful in identifying fake identities of users. Spammers can be identified based on: (i) fake content, (ii) URL based spam detection, (iii) detecting spam in trending topics, and (iv) fake user identification. Table 1 provides a comparison of existing techniques and helps users to recognize the significance and effectiveness of the

proposed methodologies in addition to providing a comparison of their goals and results. Table 2 compares different features that are used for identifying spam on Twitter. We anticipate that this survey will help readers find diverse information on spammer detection techniques at a single point.

CHAPTER-2
SYSTEM ANALYSIS

2.1 LITERATURE REVIEW

1. Authors: V. Subrahmanian and S. Kumar,

Title: ``Predicting human behavior: The next

Machine learning has provided researchers with new tools for understanding human behavior. In this article, we briefly describe some successes in predicting behaviors and describe the challenges over the next few years.

2. Authors: H. Lauw, J. C. Shafer, R. Agrawal, and A. Ntoulas

Title: ``Homophily in the digital world: A LiveJournal case study,''

Are two users more likely to be friends if they share common interests? Are two users more likely to share common interests if they're friends? The authors study the phenomenon of homophily in the digital world by answering these central questions. Unlike the physical world, the digital world doesn't impose any geographic or organizational constraints on friendships. So, although online friends might share common interests, a priori there's no reason to believe that two users with common interests are more likely to be friends. Using data from LiveJournal, the authors show that the answer to both questions is yes.

3. Authors: M. A. Al-Garadi, K. D. Varathan, and S. D. Ravana,

Title: ``Cybercrime detection in online communications: The experimental case of cyberbullying detection in the Twitter network,

The popularity of online social networks has created massive social communication among their users and this leads to a huge amount of user-generated communication data. In recent years, Cyberbullying has grown into a major problem with the growth of online communication

and social media. Cyberbullying has been recognized recently as a serious national health issue among online social network users and developing an efficient detection model holds tremendous practical significance. In this paper, we have proposed set of unique features derived from Twitter; network, activity, user, and tweet content, based on these feature, we developed a supervised machine learning solution for detecting cyberbullying in the Twitter. An evaluation demonstrates that our developed detection model based on our proposed features, achieved results with an area under the receiver-operating characteristic curve of 0.943 and an f-measure of 0.936. These results indicate that the proposed model based on these features provides a feasible solution to detecting Cyberbullying in online communication environments. Finally, we compare result obtained using our proposed features with the result obtained from two baseline features. The comparison outcomes show the significance of the proposed features.

4. Authors: L. Phillips, C. Dowling, K. Shaffer, N. Hodas, and S. Volkova, `

Title: `Using social media to predict the future:

In recent years, social media has become ubiquitous and important for social networking and content sharing. And yet, the content that is generated from these websites remains largely untapped. In this paper, we demonstrate how social media content can be used to predict real-world outcomes. In particular, we use the chatter from Twitter.com to forecast box-office revenues for movies. We show that a simple model built from the rate at which tweets are created about particular topics can outperform market-based predictors. We further demonstrate how sentiments extracted from Twitter can be further utilized to improve the forecasting power of social media.

5. Authors: H. Quan, J. Wu, and Y. Shi,

Title: ``Online social networks & social network services:

Online social network (OSN) serves as a means of social activity and has become a mainstream information media in the industry and in the public. Social network services (SNSs) have gained their popularity and serve more and more granular target markets, such as medication, science, and education. The chapter reviews mainly technology factors, as well as their psychological and social backgrounds. It deals with a background of taxonomy of OSNs by comparing similar counterparts. The chapter proposes a taxonomy and retrospectively compares it with other online communities, and even pre-internet social networking communities. It attempts to plot the possible evolution directions and future technology challenges. OSNs are used extensively as public social interactive and collaboration tools. OSNs provide various applications to facilitate social interactions and impress other people by social presence. The counteraction of profile fragmentation is studied in L. Backstrom and his team, and K. Liu and E. Terzi, as a technical challenge.

2.2 EXISTING SYSTEM

In the field of Twitter spam detection, several studies have been conducted. A few polls on false user identification from Twitter were also conducted to cover the current state-of-the-art. Present a review of new methodologies and techniques for detecting Twitter spam. The survey above provides a comparative analysis of existing techniques. The authors of conducted a survey on the various behaviours displayed by spammers on the Twitter social network. The research also includes a literature analysis that acknowledges the existence of spammers on Twitter. Despite all of the studies that have been done, there is still a void in the literature. As a result, we examine the state-of-the-art in spammer detection and fake user identification on Twitter in order to close the gap. Furthermore, this study gives a taxonomy of Twitter spam detection methods and strives to provide a comprehensive overview of current developments in the field.

2.3 PROPOSED SYSTEM

The goal of this work is to discover several ways to spam detection on Twitter and to offer a taxonomy that categorises these approaches into different groups. For the purposes of classification, we've identified four methods for reporting spammers that can assist in detecting user impersonation. Spammers can be detected using the following methods: I false content, (ii) URLbased spam detection, (iii) spam detection in popular subjects, and (iv) fake user identification. Table 1 compares existing procedures and aids users in recognising the significance and effectiveness of the proposed methodology, as well as comparing their goals and outcomes. Table 2 examines the many features used to identify spam on Twitter. We hope that by conducting this poll, readers will be able to find a wealth of information on spammer detection strategies in one place.

2.4 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- SOCIAL FEASIBLITY
-

2.4.1 ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

2.4.2 TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

2.4.3 SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system

2.5 SYSTEM REQUIREMENTS: -

2.5.1 HARDWARE REQUIREMENTS: -

- Processor - Pentium –IV
- RAM - 4 GB (min)
- Hard Disk - 20 GB
- Key Board - Standard Windows Keyboard
- Mouse - Two or Three Button Mouse
- Monitor - SVGA

2.5.2 SOFTWARE REQUIREMENTS:

- Operating System - Windows XP
- Coding Language - Java/J2EE(JSP,Servlet)
- Front End - J2EE
- Back End - MySQL

CHAPTER-3

PROJECT DESIGN

3.1 BLOCK DIAGRAM

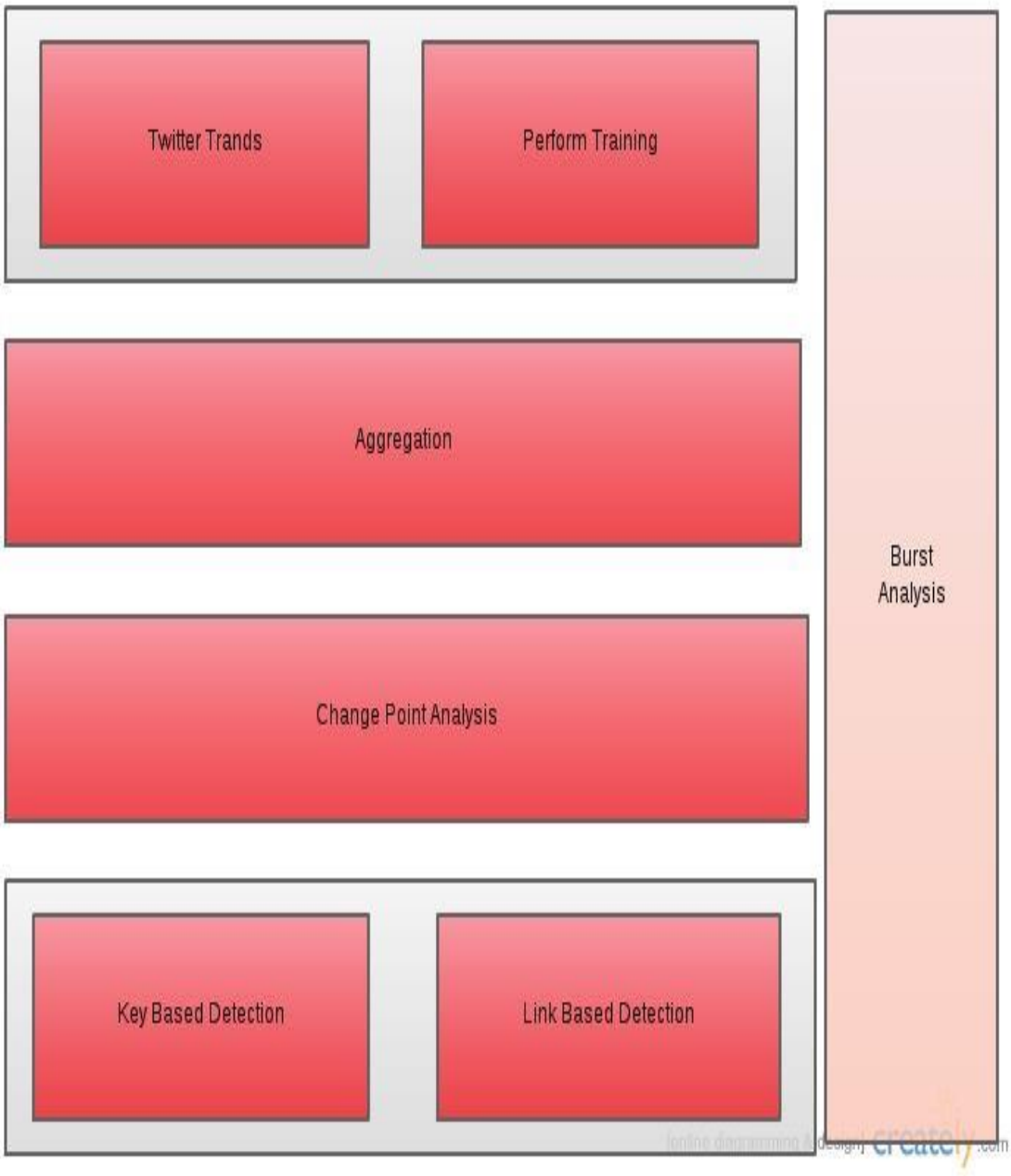


Fig.3.1: block diagram

3.2 DATA FLOW DIAGRAM:

1. The DFD is also called as bubble chart. It is a simple graphical formalism that can be used to represent a system in terms of input data to the system, various processing carried out on this data, and the output data is generated by this system.
2. The data flow diagram (DFD) is one of the most important modeling tools. It is used to model the system components. These components are the system process, the data used by the process, an external entity that interacts with the system and the information flows in the system.
3. DFD shows how the information moves through the system and how it is modified by a series of transformations. It is a graphical technique that depicts information flow and the transformations that are applied as data moves from input to output.
4. DFD is also known as bubble chart. A DFD may be used to represent a system at any level of abstraction. DFD may be partitioned into levels that represent increasing information flow and functional detail.

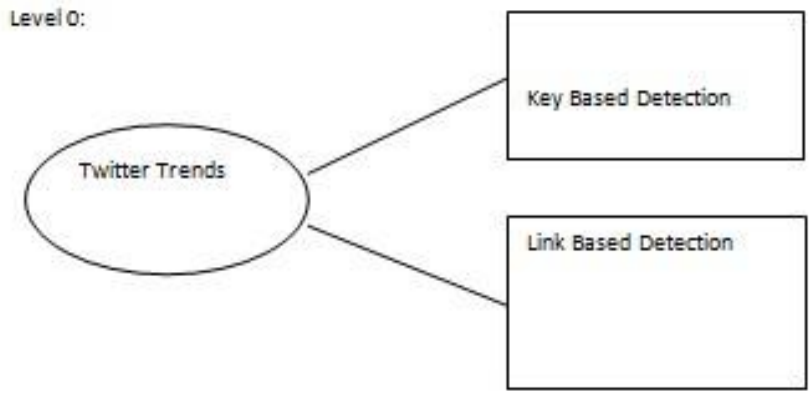


Fig.3.2.1: level 0 data flow diagram

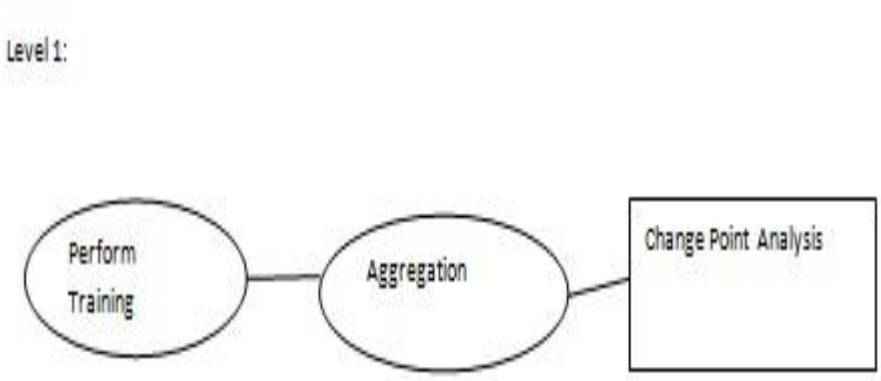


Fig.3.2.2: level 1 data flow diagram

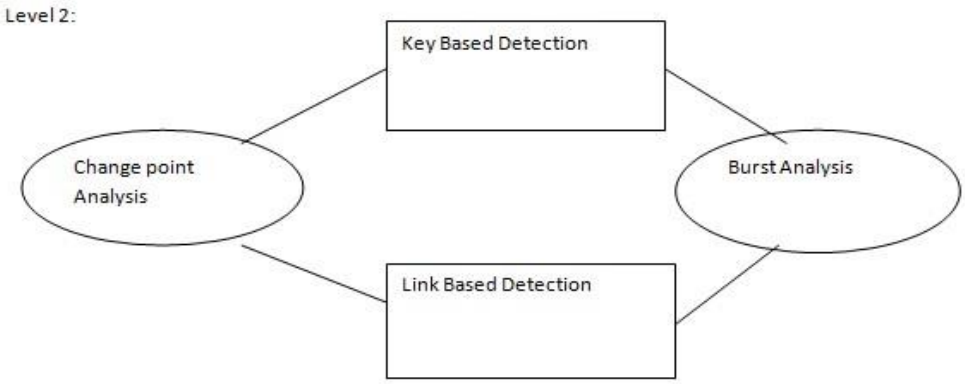


Fig.3.2.3: level 2 data flow diagram

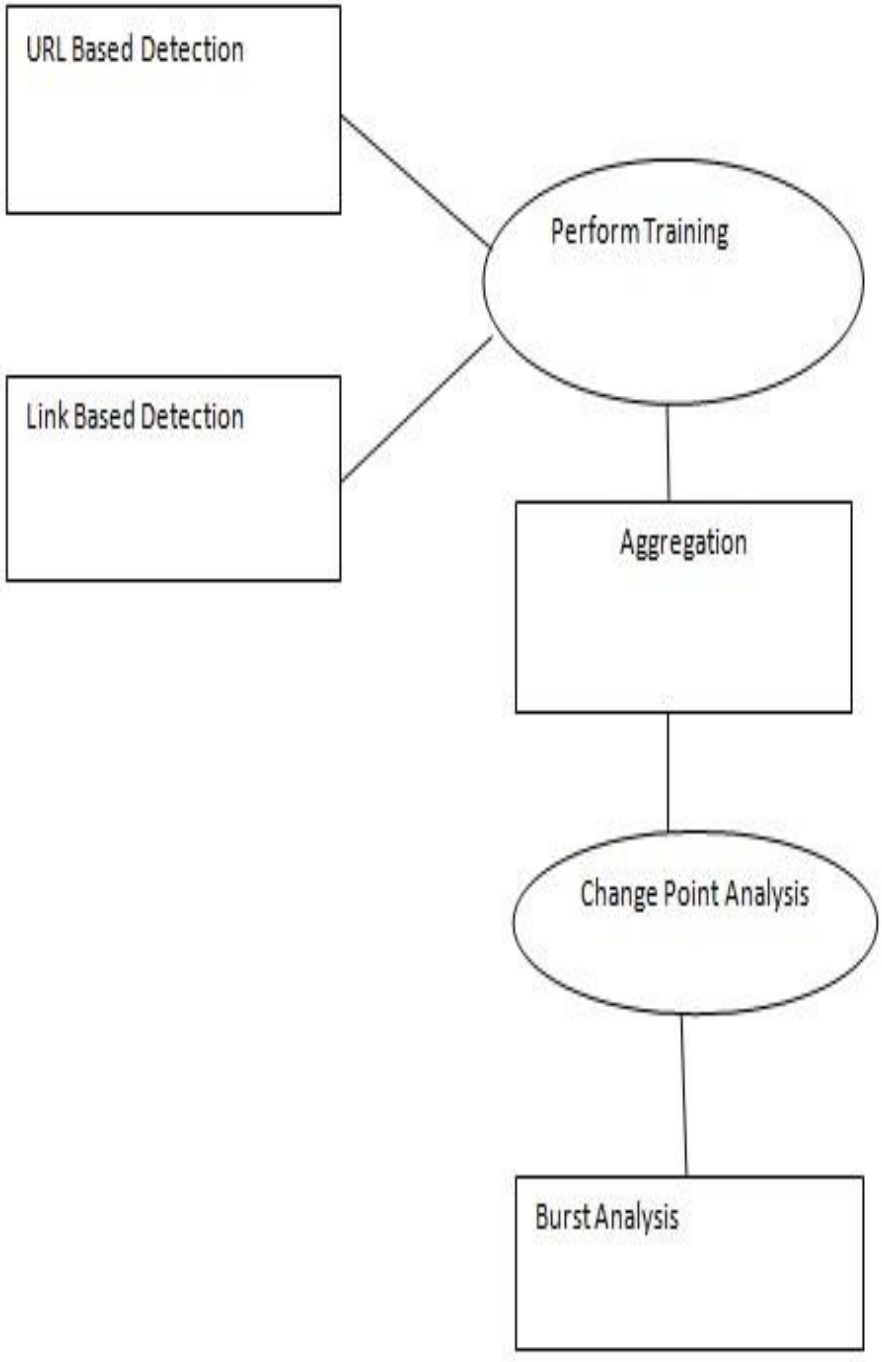


FIG.3.2.4: Workflow of the Detection and Analysis System

3.3 UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general-purpose modeling language in the field of object-oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta-model and a notation. In the future, some form of method or process may also be added to; or associated with, UML. The Unified Modelling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modelling and other non-software systems. The UML represents a collection of best engineering practices that have proven successful in the modelling of large and complex systems. The UML is a very important part of developing objects-oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

3.4 USE CASE DIAGRAM:

A use case diagram in the Unified Modelling Language (UML) is a type of behavioural diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

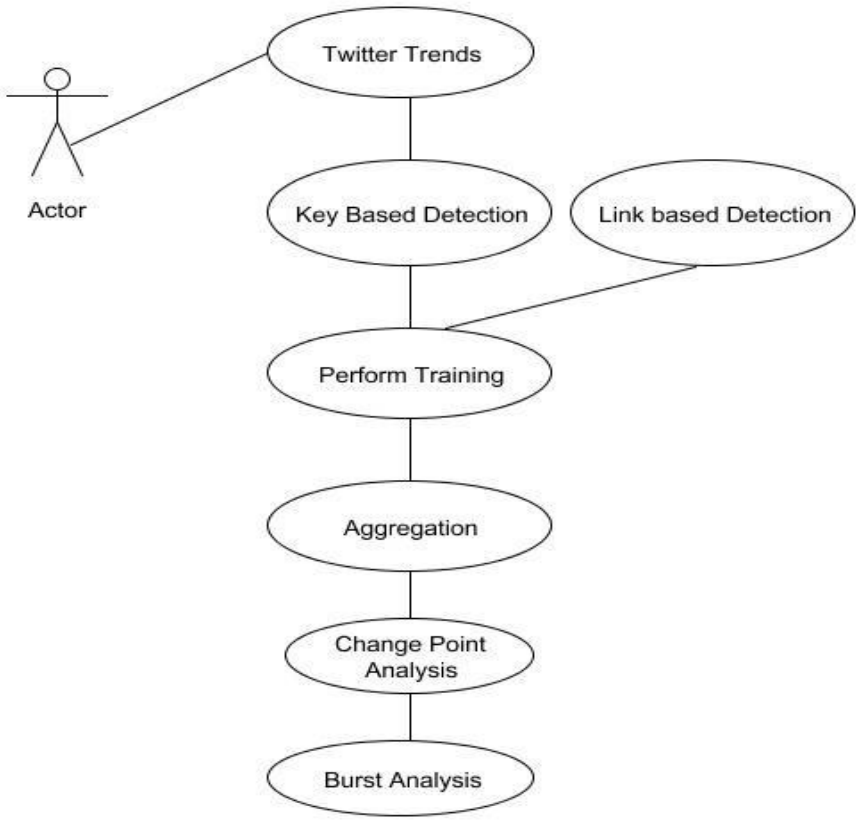


FIG.3.3: Actor-Driven Twitter Trend Detection and Analysis Framework

3.5 CLASS DIAGRAM:

In software engineering, a class diagram in the Unified Modeling Language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, operations (or methods), and the relationships among the classes. It explains which class contains information.

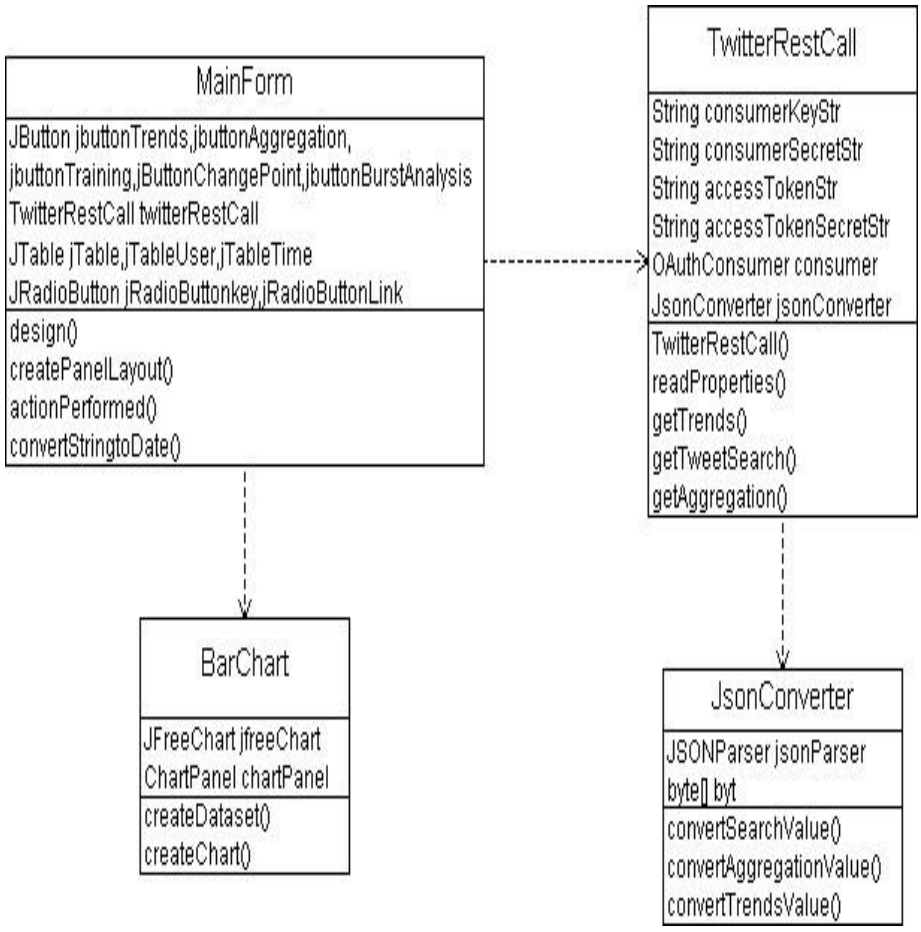


FIG.3.4: UML Class Diagram of the Twitter Data Analysis System

3.6 SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

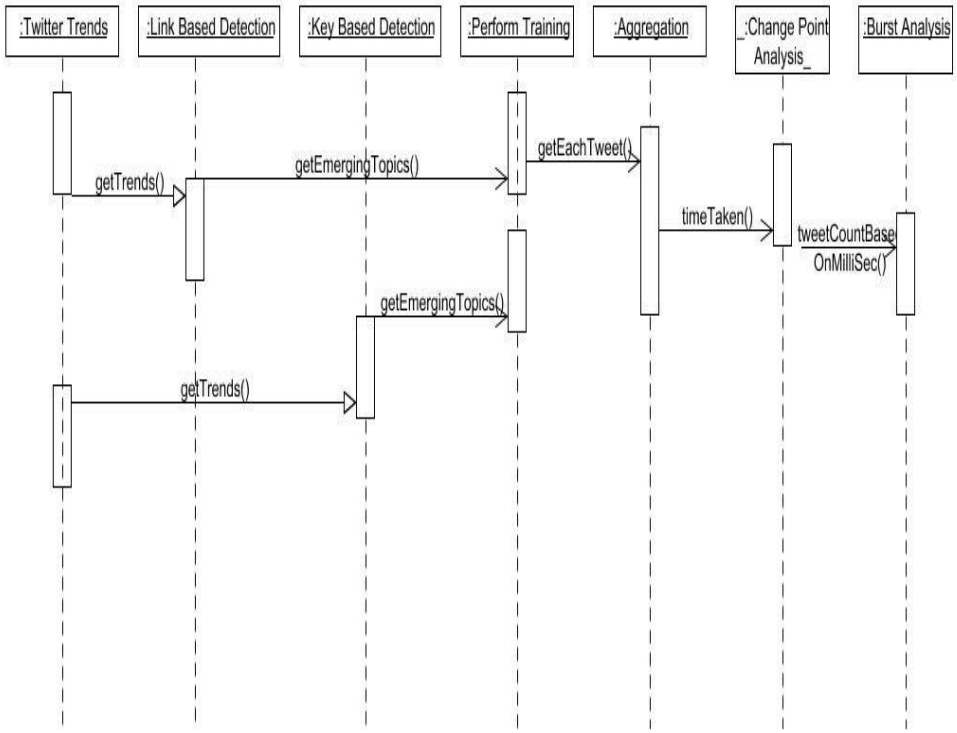


FIG.3.5: Sequence Diagram for Twitter Trend Detection and Analysis

3.7 ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modelling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control.

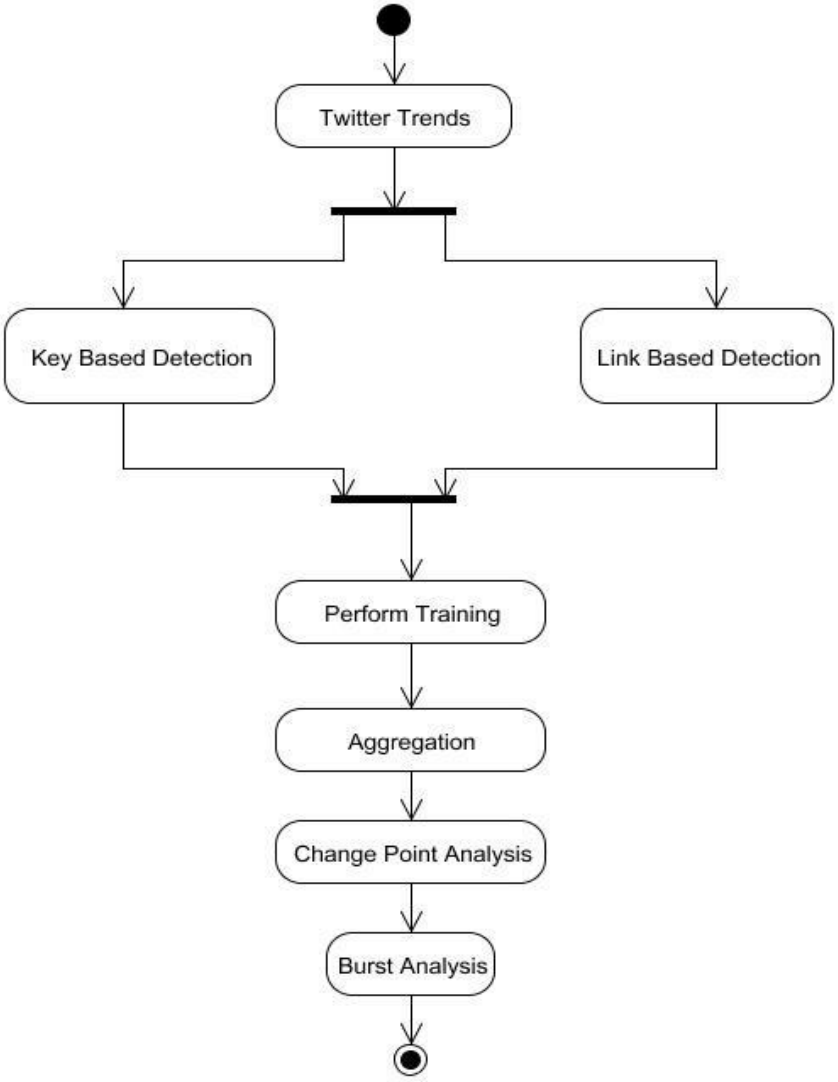


Fig.3.6: activity diagram for twitter trend detection and analysis

3.8 COLLABRATIVE DIAGRAM:

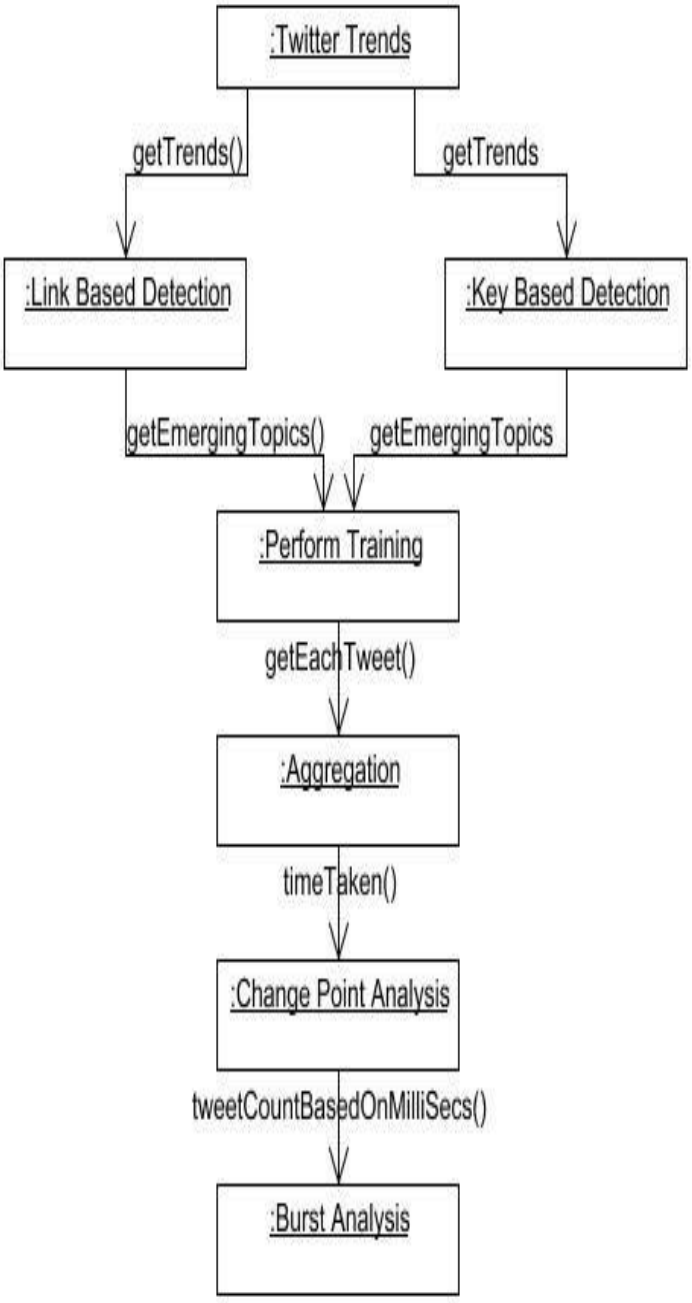


Fig.3.8:Collaborative Diagram

CHAPTER-4
IMPLEMENTATION

SOFTWARE ENVIRONMENT Java Technology

Java technology is both a programming language and a platform.

The Java Programming Language

The Java programming language is a high-level language that can be characterized by all of the following buzzwords:

- Simple
- Architecture neutral
- Object oriented
- Portable
- Distributed
- High performance
- Interpreted
- Multithreaded
- Robust
- Dynamic
- Secure

With most programming languages, you either compile or interpret a program so that you can run it on your computer. The Java programming language is unusual in that a program is both compiled and interpreted. With the compiler, first you translate a program into an intermediate language called Java byte codes —the platform-independent codes interpreted by the interpreter on the Java platform. The interpreter parses and runs each Java byte code instruction on the computer. Compilation happens just once; interpretation occurs each time the program is executed. The following figure illustrates how this works.

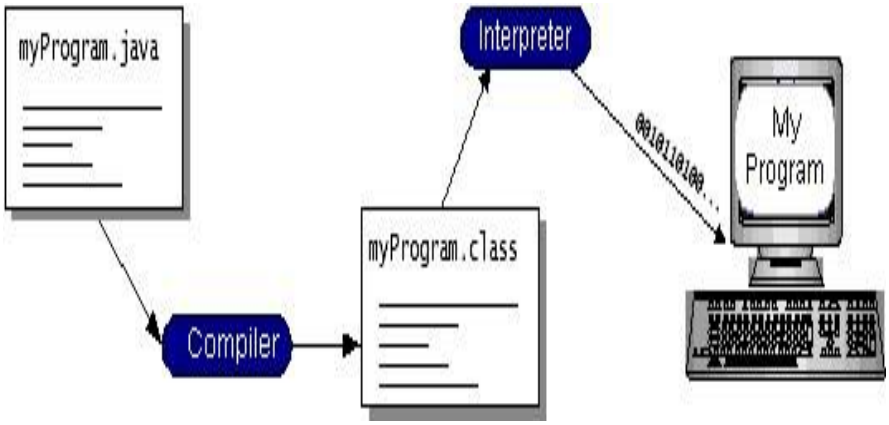


FIG.4.0: Java Program Compilation and Execution Process

You can think of Java byte codes as the machine code instructions for the Java Virtual Machine (Java VM). Every Java interpreter, whether it's a development tool or a Web browser that can run applets, is an implementation of the Java VM. Java byte codes help make "write once, run anywhere" possible. You can compile your program into byte codes on any platform that has a Java compiler. The byte codes can then be run on any implementation of the Java VM.

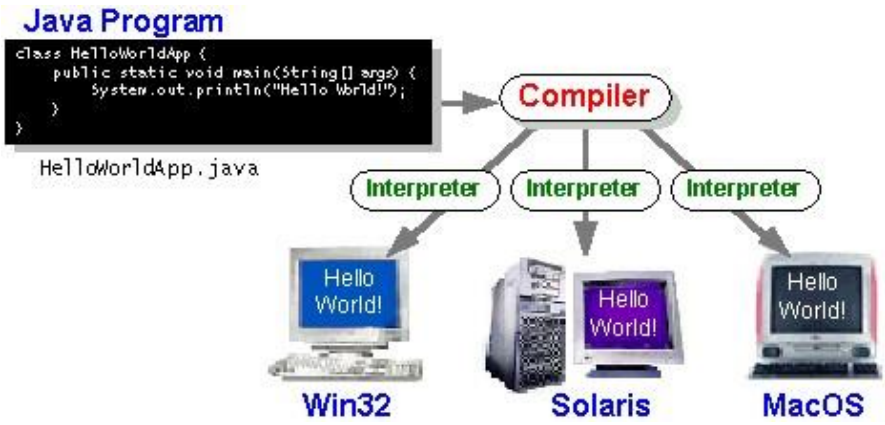


FIG.4.1: Platform Independence of Java Using the Java Virtual Machine (JVM)

The Java Platform

A platform is the hardware or software environment in which a program runs. We've already mentioned some of the most popular platforms like Windows 2000, Linux, Solaris, and MacOS. Most platforms can be described as a combination of the operating system and hardware. The Java platform differs from most other platforms in that it's a softwareonly platform that runs on top of other hardware-based platforms.

The Java platform has two components:

- The Java Virtual Machine (Java VM)
- The Java Application Programming Interface (Java API)

You've already been introduced to the Java VM. It's the base for the Java platform and is ported onto various hardware-based platforms. The Java API is a large collection of ready-made software components that provide many useful capabilities, such as graphical user interface (GUI) widgets. The Java API is grouped into libraries of related classes and interfaces; these libraries are known as packages. The next section, What Can Java Technology Do ? Highlights what functionality some of the packages in the Java API provide.

The following figure depicts a program that's running on the Java platform. As the figure shows, the Java API and the virtual machine insulate the program from the hardware.

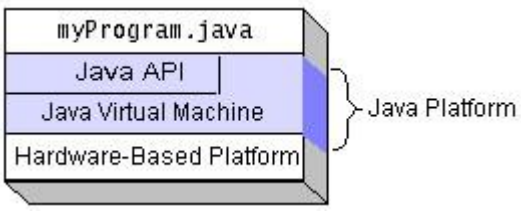


FIG.4.2: Components of the Java Platform Architecture

Native code is code that after you compile it, the compiled code runs on a specific hardware platform. As a platform-independent environment, the Java platform can be a bit slower than native code. However, smart compilers, well-tuned interpreters, and just-in-time byte code compilers can bring performance close to that of native code without threatening portability.

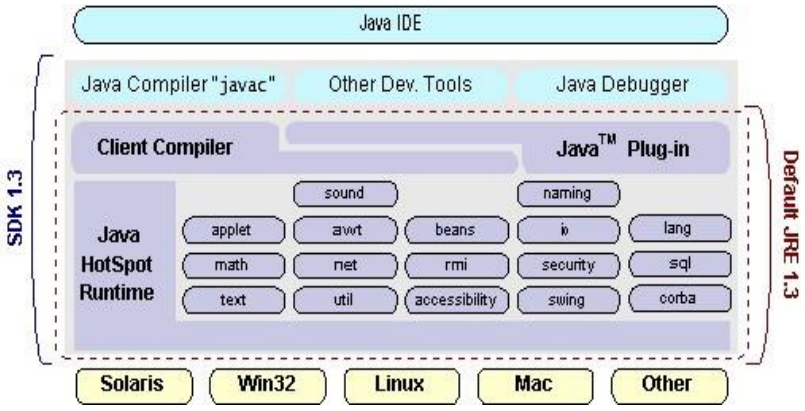


FIG.4.3: Components of the Java 2 SDK Architecture

Finally we decided to proceed the implementation using Java Networking. And for dynamically updating the cache table we go for MS Access database. Java has two things: a programming language and a platform.

Java is a high-level programming language that is all of the following

Simple	Architecture-neutral
Object-oriented	Portable
Distributed	High-performance
Interpreted	multithreaded
Robust	Dynamic
Secure	

Java is also unusual in that each Java program is both compiled and interpreted. With a compile you translate a Java program into an intermediate language called Java byte codes the platform-independent code instruction is passed and run on the computer.

Compilation happens just once; interpretation occurs each time the program is executed. The figure illustrates how this works.

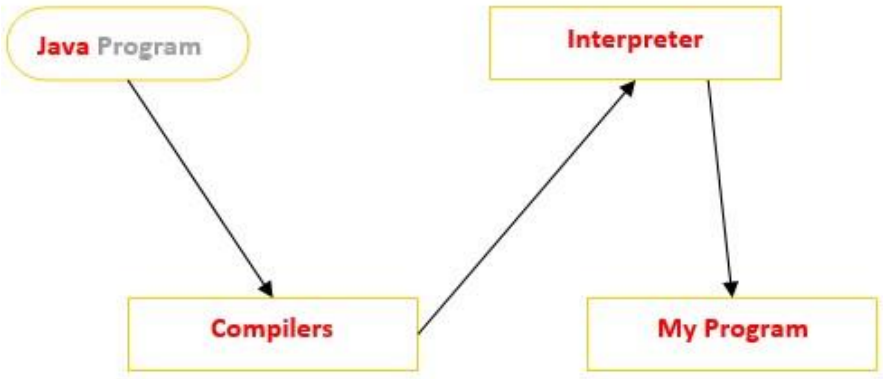


Fig.4.4:process of java program excecution using compilers & interpreter

You can think of Java byte codes as the machine code instructions for the Java Virtual Machine (Java VM). Every Java interpreter, whether it's a Java development tool or a Web browser that can run Java applets, is an implementation of the Java VM. The Java VM can also be implemented in hardware.

Java byte codes help make "write once, run anywhere" possible. You can compile your Java program into byte codes on my platform that has a Java compiler. The byte codes can then be run any implementation of the Java VM. For example, the same Java program can run Windows NT, Solaris, and Macintosh.

NETWORKING

TCP/IP stack

The TCP/IP stack is shorter than the OSI one:

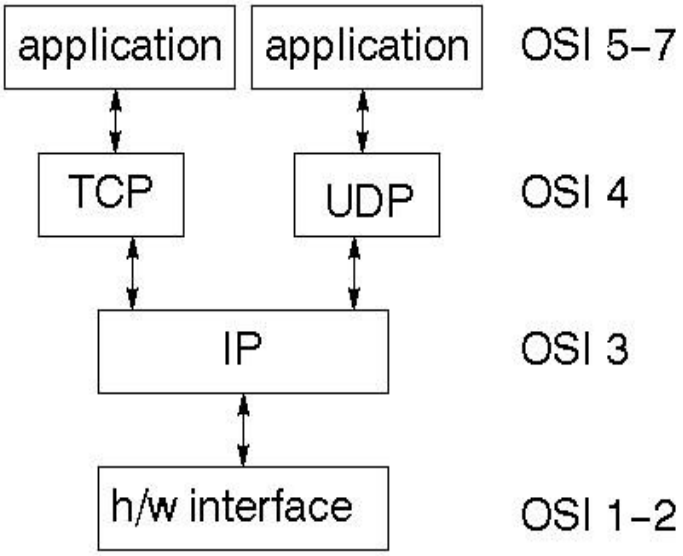


Fig.4.5: TCP/IP stack

TCP is a connection-oriented protocol; UDP (User Datagram Protocol) is a connectionless protocol.

IP datagram's

The IP layer provides a connectionless and unreliable delivery system. It considers each datagram independently of the others. Any association between datagram must be supplied by the higher layers. The IP layer supplies a checksum that includes its own header. The header includes the source and destination addresses. The IP layer handles routing through an Internet. It is also responsible for breaking up large datagram into smaller ones for transmission and reassembling them at the other end.

UDP

UDP is also connectionless and unreliable. What it adds to IP is a checksum for the contents of the datagram and port numbers. These are used to give a client/server model - see later.

TCP

TCP supplies logic to give a reliable connection-oriented protocol above IP. It provides a virtual circuit that two processes can use to communicate.

Internet addresses

In order to use a service, you must be able to find it. The Internet uses an address scheme for machines so that they can be located. The address is a 32 bit integer which gives the IP address. This encodes a network ID and more addressing. The network ID falls into various classes according to the size of the network address.

Network address

Class A uses 8 bits for the network address with 24 bits left over for other addressing. Class B uses 16 bit network addressing. Class C uses 24 bit network addressing and class D uses all 32.

Subnet address

Internally, the UNIX network is divided into sub networks. Building 11 is currently on one sub network and uses 10-bit addressing, allowing 1024 different hosts.

Host address

8 bits are finally used for host addresses within our subnet. This places a limit of 256 machines that can be on the subnet.

Total address

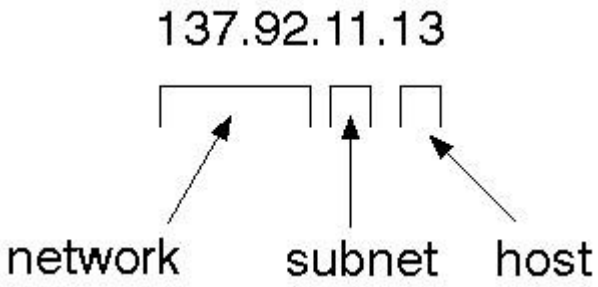


FIG.4.6: a 32-bit IP address 137.92.11.1

The 32 bit address is usually written as 4 integers separated by dots.

Port addresses

A service exists on a host, and is identified by its port. This is a 16 bit number. To send a message to a server, you send it to the port for that service of the host that it is running on. This is not location transparency!

Certain of these ports are "well known".

SAMPLE CODE CONNECT.JSP

```
<%@ page import="java.sql. *"%>
<%@ page import="java.util.*" %>
<%
Connection connection = null;
  try {
          Class.forName("com.mysql.jdbc.Driver");
          connection =
DriverManager.getConnection("jdbc:mysql://localhost:3306/myproject","
root","root");
      String sql="";
}
catch(Exception e)
{
System.out.println(e);
}
%>
```

USERREGISTER.JSP

```
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>IOT Device Reister</title>
<meta http-equiv="Content-Type" content="text/html; charset=utf-8" />
<link href="css/style.css" rel="stylesheet" type="text/css" />
<link rel="stylesheet" type="text/css" href="css/coin-slider.css" />
<script type="text/javascript" src="js/cufon-yui.js"></script>
<script type="text/javascript" src="js/cufon-aller.js"></script>
<script type="text/javascript" src="js/jquery-1.4.2.min.js"></script>
<script type="text/javascript" src="js/script.js"></script>
<script type="text/javascript" src="js/coin-slider.min.js"></script>
<script language="javascript" type="text/javascript">       function
valid()
{
var na3=document.s.userid.value;
if(na3=="")
{
alert("Please Enter IOT Device Name");
document.s.userid.focus(); return
false;
}
else
{
}
var na4=document.s.pass.value;
if(na4=="")
{
alert("Please Enter Password"); document.s.pass.focus();
return false;
}
var na9=document.s.dept.value;
if(na9=="--Select--")
{
alert("Please Select Your Department"); document.s.dept.focus();
return false;
}
var na99=document.s.spec.value;
if(na99=="--Select--")
{
alert("Please Select Your Specialist"); document.s.spec.focus();
```

```
return false;
}
var na6=document.s.email.value;
if(na6=="")
{
alert("Please Enter the Email"); document.s.email.focus();
return false;
}
if (na6.indexOf("@", 0) < 0)
{
alert("Please enter a valid e-mail address.");
document.s.T3.focus();
return false;
}
if (na6.indexOf(".", 0) < 0)
{
alert("Please enter a valid e-mail address.");
document.s.T3.focus();
return false;
}
var na7=document.s.mobile.value;
if(na7.length!=10)

{
alert("Please Enter Valid Mobile number or Enter 10 Digit number");
document.s.mobile.focus();
return false;
}
if(na7=="")
{
alert ("Please Enter Mobile number"); document.s. mobile.
Focus();
return false;
}
var na5=documents. address. Value;
if(na5=="")
{
alert ("Please Enter Your address"); document.s address. Focus
();
return false;
}
var dob=document.s.dob.value;
if(dob=="")
{
```

```
alert("please Enter Your Date Of Birth"); document.s.dob.focus();
return false;
}
var na9=document.s.gender.value;
if(na9=="--Select--")
{
alert("Please Enter Your Gender"); document.s.gender.focus();
return false;
}
var na10=document.s.pincode.value;
if(na10=="")
{
alert("please Enter Your Pincode"); document.s.pincode.focus();
return false;
}
var na11=document.s.pic.value;
if(na11=="")
{
alert("Please Select Picture"); document.s.pic.focus();
return false;
}
}
</script>
<style type="text/css">
<!--
.style1 {
        font-size: 20px;
        color: #33FF99;
}
.style2 {font-size: 25px}
.style3 {
        color: #FF0000;
        font-weight: bold;
}
-->
</style>
</head>
<body>
<div class="main">
<div class="header">
<div class="header_resize"
<div class="logo">
<h1><a href="index.html" class="style1 style2">Privacy Preserving
Medical Record Searching <br/>Scheme for IoT Healthcare</a></h1>
```

```
</div>
<div class="menu_nav">
<ul>
<li><a href="index.html"><span>Home Page</span></a></li>
<li class="active"><a href="owner_login.jsp"><span>IOT
Device</span></a></li>
<li><a href="user_login.jsp"><span>Health Care Center</span></a></li>
<li><a href="cs_login.jsp"><span>Medical Cloud
Servers</span></a></li>
<li><a href="au_login.jsp"><span>KPS</span></a></li>
</ul>
</div>
<div class="clr"></div>
<div class="slider"><div id="coin-slider"> <a href="#"><img
src="images/slide1.jpg" width="960" height="320" alt="" /> </a> <a
href="#"><img src="images/slide2.jpg" width="960" height="320" alt="" />
</a> <a href="#"><img src="images/slide3.jpg" width="960" height="320"
alt="" /> </a> </div>
<div class="clr"></div>
</div>
<div class="clr"></div>
</div>
</div>
<div class="content">
<div class="content_resize">
<div class="mainbar">
<div class="article">
<p      align="center"><img      src="images/Register.jpg"      width="343"
height="65" /></p>
<form name="s" action="ow_RegIns.jsp" method="post"
enctype="multipart/form-data" onSubmit="return valid()"  ons
target="_top">

<label for="name">
<span class="style3"><strong>IOT Device Name (required)<br />
</strong></span></label>
<p class="style3"><strong>
<input id="name" name="userid" class="text" />
</strong></p>

<span class="style3">
<label for="password">
Password (required)<br />
</label>
</span>
```

```
<p class="style3"><strong>
<input type="password" id="password" name="pass" class="text" />
 </strong></p>

 <span class="style3"><strong>
 <label for="gender"> Department (required)<br />
 </label>
 </strong></span
 <p class="style3"><strong>
 <select id="s1" name="dept" class="text" style="width:170px;">
 <option>--Select--</option>
 <option>Cardiology</option>
 <option>Neurology</option>
 <option>Nephrology</option>
 </select>
 </strong></p>
                                                    <span
class="style3"><strong>
 <label for="gender"> Specialist   (required)<br />
 </label>
 </strong></span>
 <p class="style3"><strong>
 <select id="s1" name="spec" class="text" style="width:170px;">
 <option>--Select--</option>
 <option>Heart</option>
 <option>Brain</option>
                                  <option>Kidney</option>
 </select>
 </strong></p>
                                                    <span
class="style3"><strong>
 <label for="email">Email Address (required)<br />
 </label>
 </strong></span>
 <p class="style3"><strong>
 <input id="email" name="email" class="text" />
</strong></p>
 <span class="style3"><strong>
  <label for="mobile">Mobile Number (required)<br />
  </label>
  </strong></span>
  <p class="style3"><strong>
  <input id="mobile" name="mobile" class="text" />
  </strong></p>
```

```
<span class="style3"><strong>
<label for="address">Your Address<br />
</label>
</strong></span>
<p class="style3"><strong>
<textarea name="address" cols="50" id="address"></textarea>
 </strong></p>
<span class="style3"><strong>
<label for="dob">Date of Birth (required)<br />
</label>
</strong></span>
<p class="style3"><strong>
<input id="dob" name="dob" class="text" />
</strong></p>
span class="style3"><strong>
<label for="gender">Select Gender (required)<br />
 </label>
 </strong></span>
 <p class="style3"><strong>

<select id="s1" name="gender" class="text">
<option>-Select-</option>
<option>Male</option>
<option>Female</option>
</select>
</strong></p>
<span class="style3"><strong>
<label for="pincode">Enter Pincode (required)<br />
</label>
</strong></span>
<p class="style3"><strong>
<input id="pincode" name="pincode" class="text" />
</strong></p>
<span class="style3"><strong>
<label for="pic">Select Profile Picture (required)</label>
</strong></span>
<span class="style3">
<label for="pic"></label>
</span>
<strong>
<label for="pic"></label>
</strong>
<label for="pic"><br />
</label>
```

```
 <p>
 <input type="file" id="pic" name="pic" class="text" />
  </p>
  <p>
  <input name="submit" type="submit" value="REGISTER" />
</p>
   <p align="right"><a href="owner_login.jsp">Back</a></p>
  </form>
<p> </p>
<p> </p>
</div>
</div>
 <div class="sidebar">
 <div class="gadget">
 <h2 class="star"><span>Sidebar</span> Menu</h2>
 <div class="clr"></div>
 <ul class="sb_menu">
 <li><a href="index.html"><span>Home Page</span></a></li>
 <li><a href="owner_login.jsp"><span>IOT Device</span></a></li>
 <li><a href="user_login.jsp"><span>Health Care Center</span></a>
</li>
 <li><a href="cs_login.jsp"><span>Medical Cloud Servers</span></a>
</li
<li><a href="au_login.jsp"><span>KPS</span></a></li>
</ul>
</div>
</div>
<div class="clr"></div>
</div>
</div>
<div class="fbg">
<div class="fbg_resize">
<div class="col c1">
<h2><span>Image</span> Gallery</h2>
<a href="#"><img src="images/gal1.jpg" width="75" height="75" alt=""
class="gal" /></a> <a href="#"><img src="images/gal2.jpg" width="75"
height="75" alt="" class="gal" /></a> <a href="#"><img
src="images/gal3.jpg" width="75" height="75" alt="" class="gal" /></a>
<a href="#"><img src="images/gal4.jpg" width="75" height="75" alt=""
class="gal" /></a> <a href="#"><img src="images/gal5.jpg" width="75"
height="75" alt="" class="gal" /></a> <a href="#"><img
src="images/gal6.jpg" width="75" height="75" alt="" class="gal" /></a>
</div>
<div class="clr"></div>
</div>
```

```
</div>
<div class="footer">
<div class="footer_resize">
<div style="clear:both;"></div>
</div>
</div>
</div>
<div align=center></div>
</body>
</html>
```

CHAPTER-5

RESULT

To run this project double click on 'run.bat' file to get below screen

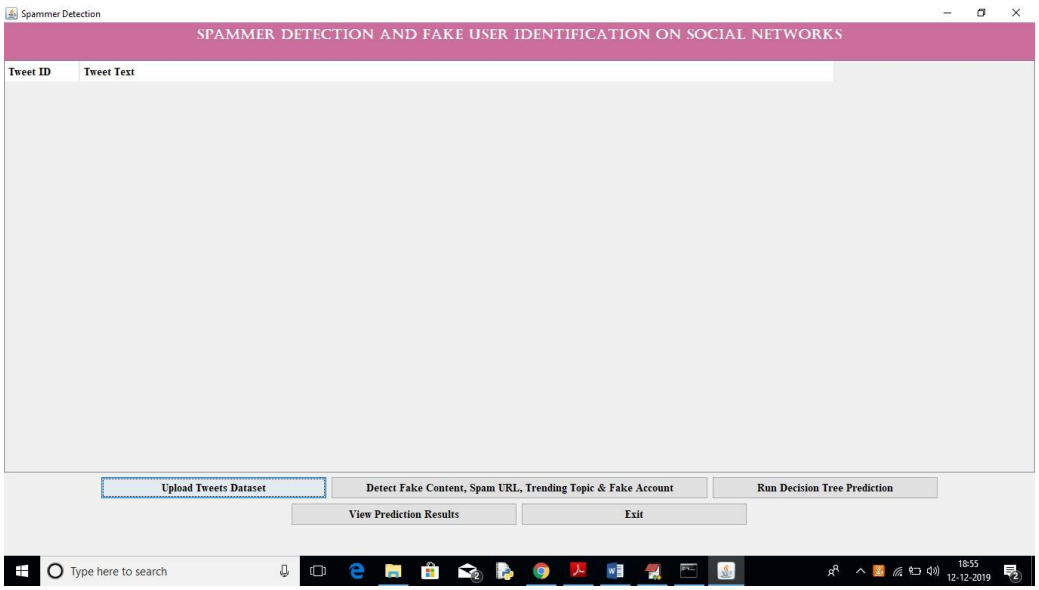


FIG.4.7: Initial Interface of Spammer and Fake User Detection System After Launching 'run.bat

In above screen click on 'Upload Tweets Dataset' button and upload Tweets

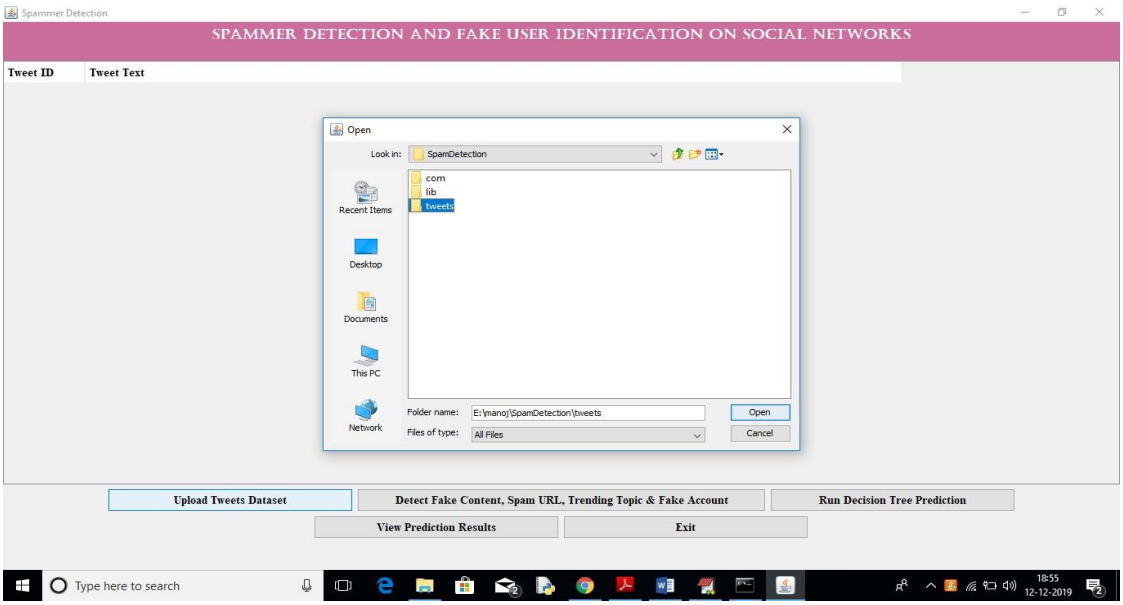


FIG.4.8: Uploading Tweets Dataset into the Spammer Detection System Interface

In above screen I am uploading 'tweets' folder which contains tweets from various users in JSON format. Now click open button to start reading tweets

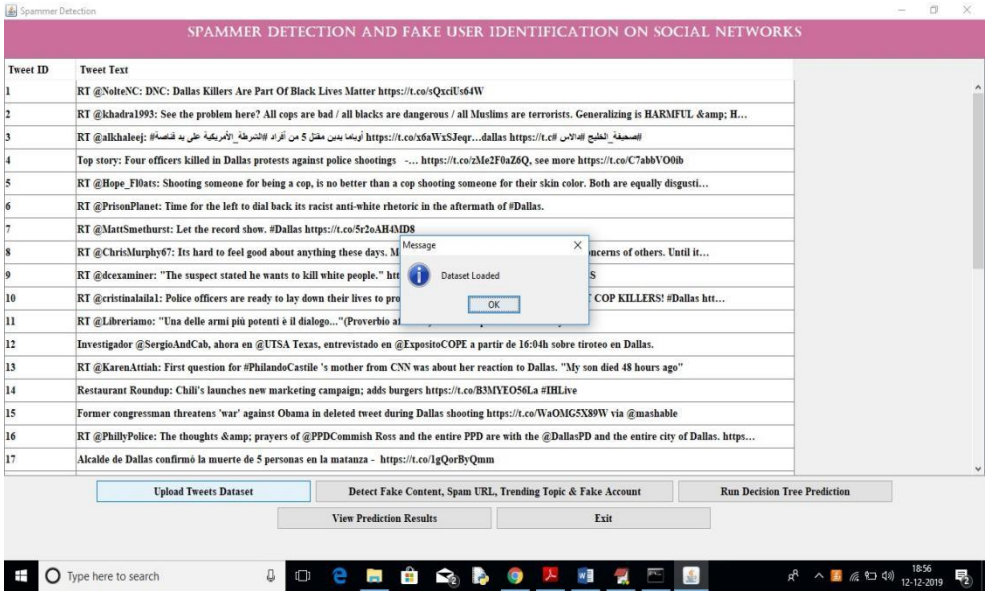


FIG.4.9: Tweets Dataset Successfully Loaded into the Spammer Detection System Interface

In above screen we can see all tweets from all users. First column contains user id and second column contain tweets from that user. Now click on 'Detect Fake Content, Spam URL, Trending Topic & Fake Account' button to analyse all tweets with 4 techniques and below are the detection results

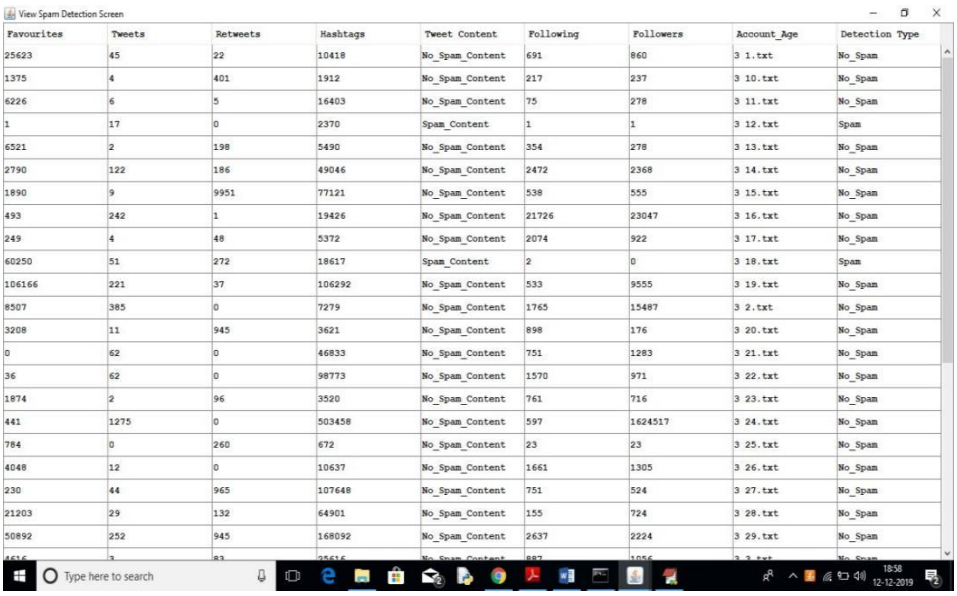


FIG.5.0: Detection Results Displayed After Analyzing Tweets Using Spam and Fake Account Identification Techniques

In above screen all features extracted from tweets dataset and then analyse those features to identify tweets is no spam or spam. In the last column we can see detection result and each spam row has less number of followers and following which indicates this account is fake and user is using this account just to spread spam messages and he is not making any friend or following anybody. Now click on 'Run Random Forest Prediction' button to classify/predict all records.

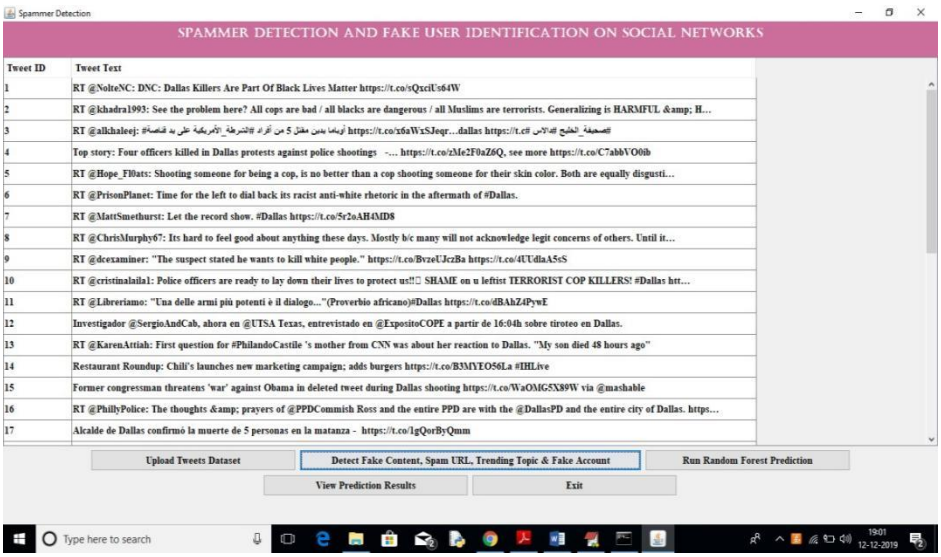


FIG.5.1: Tweet Data Displayed Before Running Random Forest Prediction for Classification

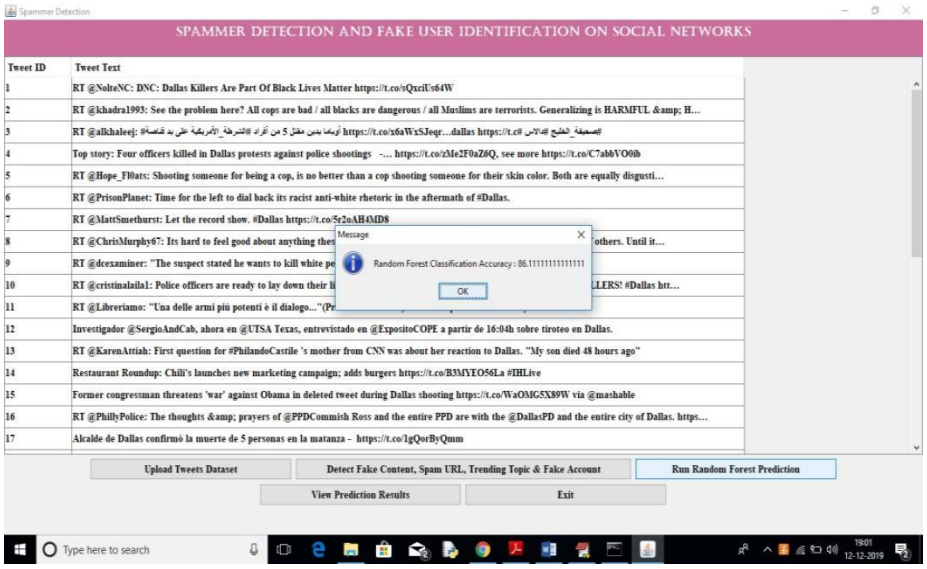


FIG.5.2: Random Forest Classification Completed with Accuracy Displayed in Spammer Detection System

above screen we got random forest prediction accuracy as 86%, now click on 'View Prediction Results' button to view number of predicted spam and no spam tweets.

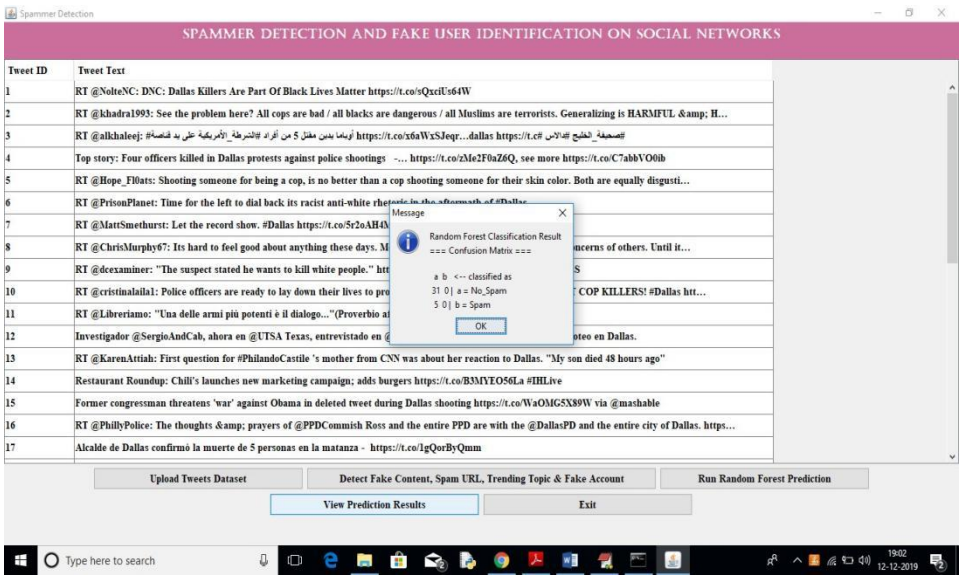


FIG.5.3: Random Forest Classification Results with Confusion Matrix Displayed for Spam Detection

In above screen we can see number of no spam predicted records are 31 and spam predicted records are 5

CHAPTER-6
TESTING

6.1 SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

6.2 TYPES OF TESTS

6.2.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

6.2.2 Integration testing:

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

6.2.3 Functional testing:

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.
Functional testing is centered on the following items:
Valid Input : identified classes of valid input must be accepted.
Invalid Input : identified classes of invalid input must be rejected.
Function : identified functions must be exercised.
Output : identified classes of application outputs must be exercised.

6.2.4 System Testing:

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

6.2.5 White Box Testing:

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

6.2.6 Black Box Testing:

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing:

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

6.3 Test strategy and approach:

Field testing will be performed manually and functional tests will be written in detail.

Integration Testing:

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error

Acceptance Testing:

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results:

All the test cases mentioned above passed successfully. No defects encountered.

CHAPTER-7
CONCLUSION

We conducted a review of approaches for detecting spammers on Twitter in this research. Furthermore, we proposed a taxonomy of Twitter spam detection strategies, dividing them into four categories: fake content identification, URL based spam detection, spam identification in hot topics, and false user detection strategies. We also examined the offered strategies based on a variety of factors, including user characteristics, content characteristics, graph characteristics, structural characteristics, and temporal characteristics. Furthermore, the strategies were compared in terms of the aims they were designed to achieve and the datasets they employed. The given review is expected to aid academics by providing a comprehensive source of information on state-of-the-art Twitter spam detection systems.

FUTURE SCOPE

Despite the development of efficient and successful ways for spam detection and fake user identification on Twitter , there are still certain gaps in the study that need to be addressed. The following are a few of the issues: Because of the substantial ramifications of false news on an individual and communal level, false news identification on social media networks is a subject that needs to be investigated. The identification of rumour origins on social media is another related topic worth researching. Although a few studies using statistical methods to discover the origin of rumours have already been undertaken, more complex approaches, such as social network-based approaches, can be used due to their demonstrated efficiency.

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